

Task

Write a 5-paragraph short essay on "AI will replace humans altogether".

Discuss the extent to which you agree or disagree with the statement and explain your reasoning for the position you take. In developing and supporting your position, you should consider ways in which the statement might or might not hold true and explain how those considerations shape your position.

A simple guideline for structuring your essay can be found here:

<https://www.kaptest.com/study/gre/how-to-structure-the-gre-issue-essay/>

Submission

You need to submit a PDF file: 'StudentID_q4.pdf' (StudentID will be replaced by your student ID) containing your essay.

Instructions

- Read into the issue, formulate your own opinion on it, and then articulate and develop an argument to support your opinion on the issue
- The essay should consist of 500-600 words providing a solid line of reasoning and examples.
- Late submissions and plagiarism will be heavily penalized.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 3**DURATION: 20 Minutes****WINTER SEMESTER, 2021-2022****FULL MARKS: 15**

CSE 4711: Artificial Intelligence

Answer all **2 (two)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

Student ID: _____

1. In the game tree shown in Figure 1, an up triangle ($\triangle$) represents a maximizer node, a down triangle (∇) represents a minimizer node, a circle ($\circ$) represents a chance node, and a square ($\square$) represents a terminal node. The value beside the arcs indicate the probability of taking the corresponding action. In the event of a tie, the left action is taken. Here, P and Q are whole numbers.

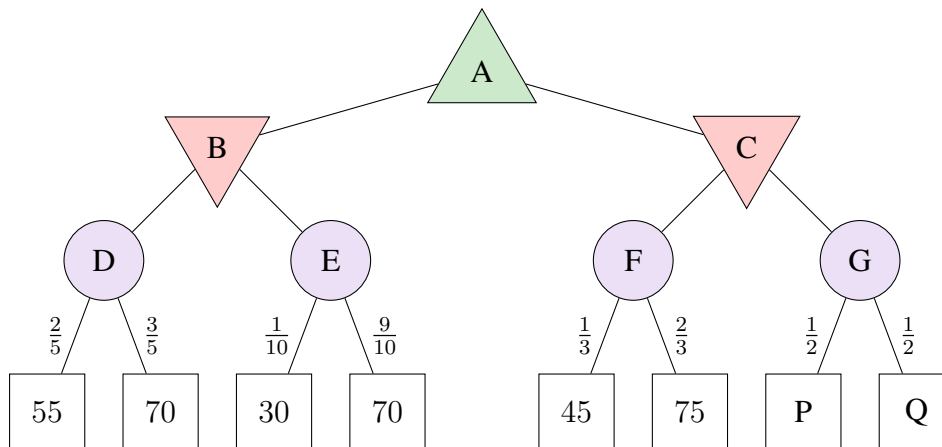


Figure 1: Game Tree for Question 1

Assume that the players play optimally. Now answer the following questions:

- a) Determine the value of the nodes that do not depend on P and Q.

Solution: $V(D) = \frac{2}{5} \times 55 + \frac{3}{5} \times 70 = 64$
 $V(E) = \frac{1}{10} \times 30 + \frac{9}{10} \times 70 = 66$
 $V(F) = \frac{1}{3} \times 45 + \frac{2}{3} \times 75 = 65$
 $V(B) = \min(64, 66) = 64$

Rubric:

- 1 point for each value.

4
(CO1)
(PO1)

- b) With a brief explanation, determine the condition(s) that P and Q must satisfy for the following scenarios:

- i. The game will progress towards C.

4 + 5
(CO1)
(PO1)

Solution: For A to pick C over B,

$$\begin{aligned}V(C) &> V(B) \\ \Rightarrow \min(V(F), V(G)) &> 64 \\ \Rightarrow \min\left(65, \frac{P+Q}{2}\right) &> 64 \\ \Rightarrow \frac{P+Q}{2} &> 64 \\ \Rightarrow P+Q &> 128\end{aligned}$$

Rubric:

- 1 point for the correct initial relation
- 2 points for the steps
- 1 point for the correct lower bound

ii. The game will progress towards G.

Solution: From the previous question, to get to C, we need $P+Q > 128$.
For C to pick G over F,

$$\begin{aligned}V(G) &< V(F) \\ \Rightarrow \frac{P+Q}{2} &< 65 \\ \Rightarrow P+Q &< 130\end{aligned}$$

Combining the two equations, we get: $128 < P+Q < 130$.
Since P and Q are whole numbers, $P+Q = 129$.

Rubric:

- 1 point for the correct initial relation
- 2 points for the steps
- 1 point for the correct upper bound
- 1 point for the exact value

2. A meme is an image, video, piece of text, etc., typically humorous in nature, that is copied and spread rapidly by internet users, often with slight variations. 2

Create a meme related to this course. A text description will suffice.

Solution:



imgflip.com

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QUIZ 2**DURATION: 20 Minutes****WINTER SEMESTER, 2021-2022****FULL MARKS: 15**

CSE 4711: Artificial Intelligence

Answer all **3 (three)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

Student ID: _____

Figure 1 shows the state space graph for an informed search problem. Here, A is the start state and G is the goal state. The cost for executing each action (in both directions) is shown near each arc.

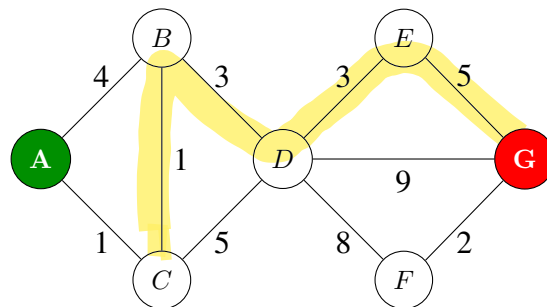


Figure 1: State Space Graph

An incomplete heuristic function, h is shown in Table 1.

Table 1: Incomplete Heuristic Function

Node	A	B	C	D	E	F	G
h	10	9	?	7	4.5	1.5	0

1. For each of the following scenarios, determine the tightest upper bound and lower bound for $h(C)$ with brief explanation. **(CO1, PO1)**

a) h is admissible.

4

Solution: To make h admissible, $h(C)$ has to be less than or equal to the actual optimal cost from C to goal G , which is the cost of the path $C \rightarrow B \rightarrow D \rightarrow E \rightarrow G$, i.e. 12. And the minimum possible value for heuristic is no heuristic, which makes all the values 0. The answer is $0 \leq h(C) \leq 12$.

Rubric:

- For the correct range: 2 points
- For a correct explanation of the lower bound: 1 point
- For a correct explanation of the upper bound: 1 point

b) h is consistent.

5

Solution: All the other nodes except node C satisfy the consistency conditions. The consistency conditions that involve the state C are:

$$\bullet h(A) \leq c(A, C) + h(C)$$

$$\rightarrow 10 \leq 1 + h(C)$$

$$\rightarrow 9 \leq h(C)$$

$$\bullet h(B) \leq c(B, C) + h(C)$$

$$\rightarrow 9 \leq 1 + h(C)$$

$$\rightarrow 8 \leq h(C)$$

$$\bullet h(D) \leq c(D, C) + h(C)$$

$$\rightarrow 7 \leq 5 + h(C)$$

$$\rightarrow 2 \leq h(C)$$

All goes through C

$$\bullet h(C) \leq c(C, A) + h(A)$$

$$\rightarrow h(C) \leq 1 + 10$$

$$\rightarrow h(C) \leq 11$$

$$\bullet h(C) \leq c(C, B) + h(B)$$

$$\rightarrow h(C) \leq 1 + 9$$

$$\rightarrow h(C) \leq 10$$

$$\bullet h(C) \leq c(C, D) + h(D)$$

$$\rightarrow h(C) \leq 5 + 7$$

$$\rightarrow h(C) \leq 12$$

C goes through all

Using the tightest bounds from the equations above: $9 \leq h(C) \leq 10$

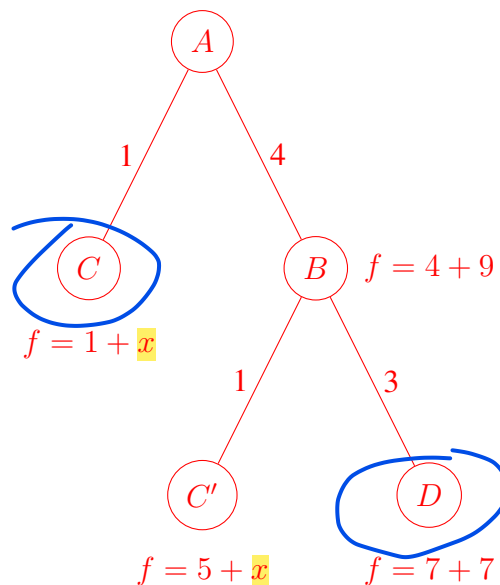
Rubric:

- For the correct range: 2 points
- For a correct explanation of the lower bound: 1.5 point
- For a correct explanation of the upper bound: 1.5 point

c) A* Graph Search expands nodes in the following order: $A \rightarrow B \rightarrow C \rightarrow D$.

6

Solution: Let, $h(C) = x$.



The search tree for A* Graph Search using the heuristic h is shown above. Node A is expanded first since it's the start state. After expanding A , B and C will go into the fringe.

To make A* graph search expand node B before node C , we need

$$f(C) > f(B)$$

$$1 + x > 13$$

$$x > 12$$

After expanding B , C and D will go into the fringe. We denote this C as C' . Now, we need to either expand C' or C before D . To expand C' ,

$$F(C') < F(D)$$

$$5 + x < 14$$

$$x < 9$$

which violates the earlier constraint. So we cannot expand C' . To expand C ,

$$F(C) < F(D)$$

$$1 + x < 14$$

$$x < 13$$

After expanding C , C' and D will be in the fringe. Here, $F(D) = 14$ and $F(C') > 5 + 12 = 17$ (Considering the lower bound). So, D will be expanded afterwards.

So we can get $12 < h(C) < 13$.

Rubric:

- For the correct range: 2 points
- For a correct explanation of the lower bound: 2 points
- For a correct explanation of the upper bound: 2 points

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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Department of Computer Science and Engineering (CSE)

QUIZ 2**DURATION: 20 Minutes****WINTER SEMESTER, 2021-2022****FULL MARKS: 15****CSE 4711: Artificial Intelligence**

Answer all **2 (two)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

Student ID: _____

The game SurvivalMan is played in a grid consisting of cells. The protagonist, Pacman, and the antagonist, Ghost take turns moving horizontally or vertically given it does not go beyond the grid. In each turn, the player (Pacman/Ghost) must move to an adjacent cell. **Each move requires 2 seconds.** The game ends when Ghost catches Pacman. The terminal payoff for Pacman is the amount of time survived.

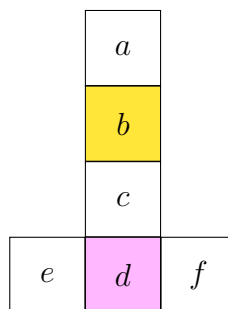
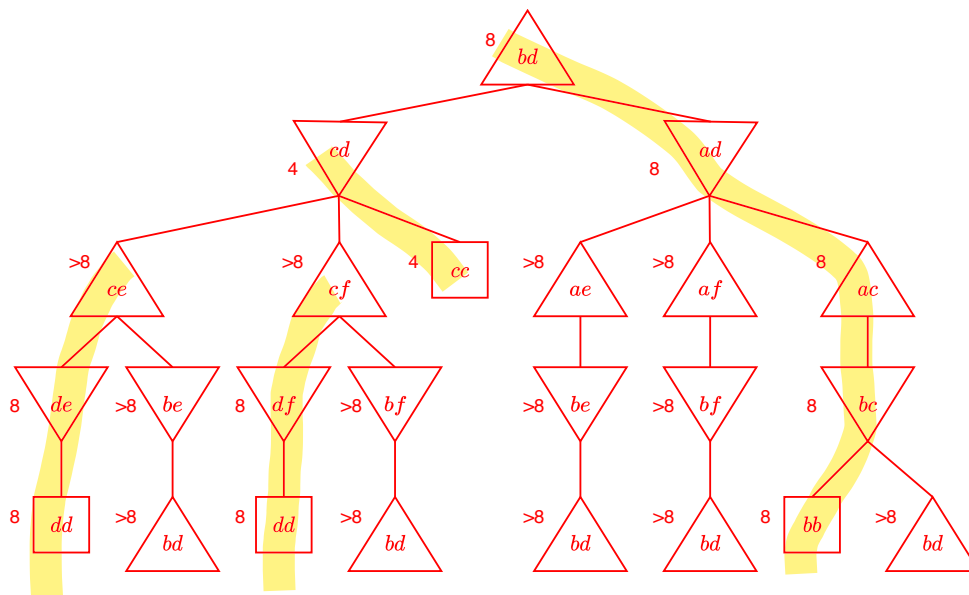


Figure 1: Game Map of SurvivalMan for Questions 1 and 2

Consider that a game of SurvivalMan is being played on the grid shown in Figure 1. Initially, Pacman is in cell *b*, Ghost is in cell *d*, and Pacman moves first. As a player controlling Pacman, your target is to evade Ghost as long as possible.

1. Draw the Minimax Tree for SurvivalMan up to depth 4 (maximum two moves for Pacman, and maximum two moves for Ghost). Use up triangles ($\triangle$) for maximizer nodes, down triangles (∇) for minimizer nodes, and squares ($\square$) for terminal nodes. Name each node using a pair of values, where the first value denotes the position of Pacman, and the second value denotes the position of Ghost at that state. You also need to show the value for each state. 12
(CO2)
(PO2)

Solution:



Rubric:

- 0.5 points for each node (should have the correct symbol, value and name).

2. With a brief explanation, determine the maximum score you can get in an optimal play.

3
(CO3)
(PO2)

Solution: From the top value of the Minimax Tree, the answer is 8.

Alternate Solution: From the Game Map, it can be seen that the best Pacman can do is to run away from Ghost. An optimal Ghost will go toward Pacman, and eventually catch it. In that case, it requires 4 steps to catch Pacman. Since each step requires 2 seconds, the maximum score should be $4 \times 2 = 8$.

Rubric:

- 1 point for the correct answer.
- 2 points for the explanation.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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QUIZ 1**DURATION: 20 Minutes****WINTER SEMESTER, 2021-2022****FULL MARKS: 15****CSE 4711: Artificial Intelligence**

Answer all **2 (two)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

Student ID: _____

The Missionaries and Cannibals problem goes as follows: **three missionaries** and **three cannibals** are on one side of a river, along with a boat that can hold one or two people. The **goal is to take all the missionaries and cannibals to the other side of the river**. In one move, **one or two people can use the boat (given it is on their side) to go the other side of the river**. However, on any side, the **cannibals must not outnumber the missionaries (if they do, the cannibals will eat the missionaries)**.

We want to formulate this scenario as a search problem. Given the missionaries, cannibals, and a boat are on one side of the river, our goal is to apply a search algorithm to find out the plan to take them to the other side.

1. Model the state space graph for this problem. Your model should describe the states, actions available, successor function, start state, and goal test. 10
(CO2, PO4)

Solution: State: Six-tuple of integers listing the number of missionaries, cannibals, and boats on each side of the river. E.g., $[M_{left}, C_{left}, B_{left}, M_{right}, C_{right}, B_{right}]$

Actions: Given the boat exists on one side of the river, 1/2 person can be moved to the other side.

Successor Function: Choose the action that move 1 or 2 people and 1 boat from one side to another. The cost can be the number of moves.

Start State: $[3, 3, 1, 0, 0, 0]$

Goal Test: The state should be $[0, 0, 0, 1, 3, 3]$

Rubric:

- 2 points for each portion

2. Considering both the tree search and graph search variants of Breadth-First Search (BFS) and Depth-First Search (DFS), which algorithm will you prefer to find the solution? Justify your answer. 2 + 3

(CO3, PO2)

Solution: We should use BFS graph search here. From all but the first and last states, there is only one other choice. As most of the actions would return us to a previously visited state, graph search can help avoid repeating states. **Since the state space is very small, we can use BFS to find the optimal solution quickly.** Alternatively, DFS can also be used to find the optimal solution, considering single valid choice per state.

Rubric:

- 1 + 1 points for correct choice (Variant + Algorithm).
- 2 + 1 points for reasoning (Variant + Algorithm).

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QUIZ 1**DURATION: 20 Minutes****WINTER SEMESTER, 2021-2022****FULL MARKS: 15****CSE 4711: Artificial Intelligence**

Answer all **2 (two)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

Student ID: _____

A famous conjecture from Donald E. Knuth states that starting with the number 4, a sequence of factorial, square root, and floor operations can be performed to reach any desired positive integer. For example, we can reach 5 from 4 using:

$$\left\lfloor \sqrt{\sqrt{\sqrt{\sqrt{\sqrt{(4!)!}}}}} \right\rfloor = 5$$

We want to formulate this scenario as a search problem. Given any number, our goal is to apply a search algorithm to find out whether we can start from 4 to reach that number.

1. Model the state space graph for this problem. Your model should describe the states, actions available, successor function, start state, and goal test. 10
(CO2, PO4)

Solution: **State:** Positive numbers. Since there are infinite positive numbers, there can be infinite number of states.

Actions: 3 actions: factorial, square root, and floor, if the current state is an integer. 2 actions: square root and floor, if the current state is not an integer.

Successor Function: Apply one of the available operations on the current state. There will be an edge connecting the states. The cost can be the number of operations.

Start State: The number 4.

Goal Test: State is the desired positive integer.

Rubric:

- 2 points for each portion

2. Considering both the tree search and graph search variants of Breadth-First Search (BFS) and Depth-First Search (DFS), which algorithm will you prefer to find the solution? Justify your answer. 2 + 3

(CO3, PO2)

Solution: We should use BFS graph search. Since the state space is infinite, the search tree can be huge. To ensure the optimal solution is found quickly, we should use BFS graph search. Using graph search can result in small number of nodes being expanded.

Rubric:

- 1 + 1 points for correct choice (Variant + Algorithm).
- 2 + 1 points for reasoning (Variant + Algorithm).

X

DFS

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MID-SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2021-2022
FULL MARKS: 75

CSE 4711: Artificial Intelligence

Answer **all 3 (three)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. Consider the game of Pacman, where the main protagonist Pacman has to move through an $M \times N$ grid eliminating ghosts. Pacman starts from an empty square and can choose one of the four following actions: left, right, up, and down to move to another square, if it does not contain a wall or go beyond the board. Pacman can eliminate a ghost by moving to the same square as the ghost. The goal is to eliminate all the G ghosts present on the board.

For simplicity, we assume that the ghosts are stationary and the cost of executing each action is 1.

- a) Determine the size of the minimal state space graph.

1
(CO1)
(PO1)

Solution: $O(M \times N \times 2^G)$

Pacman can be in one of the $M \times N$ squares.

For each square, there can either be a ghost or no ghost, which gives a total 2^G possibilities.

Rubric:

- 0.5 points for the exact value.
- 0.5 points for explanation.

- b) With proper justification, determine whether each of the following heuristics is only admissible, only consistent, none of them, or both of them:

4×6
(CO3)
(PO2)

- i. The sum of the Manhattan Distance from Pacman to every ghost.

Solution: None of them.

Consider the example where all the ghosts are adjacent and lined up horizontally. If there are three ghosts and Pacman is on their left, the sum of the Manhattan Distance will be $1 + 2 + 3 = 6$, which is an overestimate of the actual cost of 3, since Pacman only needs to move three spaces to eliminate all the ghosts.

- 1 point for the correct choice.
- 5 points for justification.

- ii. The number of remaining ghosts.

Solution: Both of them.

In each turn, Pacman can only eliminate at most one ghost, and therefore the number of ghosts will be a lower bound on the number of moves necessary to eliminate all the ghosts, making this heuristic admissible. Additionally, since between moves, the number of ghosts can only be reduced by at most one, the heuristic is also consistent.

- 1 point for the correct choice.

- 5 points for justification.

iii. The minimum Manhattan Distance between Pacman and any of the ghosts.

Solution: Both of them.

Pacman, at the very least, needs to cross the minimum of the distances between Pacman and the ghosts to reach the closest ghost. For this reason, the heuristic is consistent, which also implies admissibility

- 1 point for the correct choice.
- 5 points for justification.

iv. The number of remaining ghosts multiplied by the minimum Manhattan Distance between Pacman and any of the ghosts.

Solution: None of them.

Consider the case where Pacman is very far from the closest ghost, but all the ghosts are next to each other. The heuristic will give us an overestimate to collect all the ghosts because going the minimum distance value to each ghost is an overestimate once you have reached the closest ghost.

- 1 point for the correct choice.
- 5 points for justification.

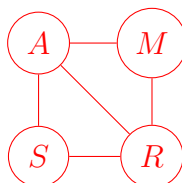
2. a) Four people, Ali (A), Sristy (S), Maliha (M), and Rafid (R) are looking to rent space in an apartment building. There are three floors in the building: 1, 2, and 3 (where 1 is the lowest and 3 is the highest). More than one person can live on a single floor, but each person must be assigned to some floor. The following constraints must be satisfied on assignment:

- A and S must not live on the same floor.
- If A and M live on the same floor, they must be on floor 2.
- If A and M live on different floors, one of them must be on floor 3.
- R must not live on the same floor as anyone else.
- R must live on a higher floor than M .

We formulate the scenario as Constraint Satisfaction Problem (CSP) considering each person as a variable and the floors as values in their domains.

i. Sketch the constraint graph for the CSP.

Solution:



Rubric:

- 0.5 points for each node

5
(CO2)
(PO2)

- 0.5 points for each correctly included arc
- 0.5 points for each correctly omitted arc

ii. Apply arc consistency to determine the remaining values in the domain of each variable.

9
(CO3)
(PO2)

Solution: $A = \{2, 3\}$

$S = \{1, 2, 3\}$

$M = \{1, 2\}$

$R = \{2, 3\}$

3 is removed from M because, for that assignment R will not be able to live above M .

1 is removed from R because, for that assignment M will not be able to live below R .

1 is removed from A because, for that assignment M must live in 3, which has been removed previously.

Rubric:

- 0.5 points for each correctly included value
- 0.5 points for each correctly omitted value
- 1 point for each explanation

iii. Let's assume A chooses to live on floor 3. Solve the CSP to determine which person lives on which floor.

7
(CO3)
(PO1)

Solution: $A = 3$

$S = 1$

$M = 1$

$R = 2$

If A stays on 3, R cannot be on the same floor. So, $R = 2$.

If R stays on 2, M needs to stay below. So, $M = 1$.

If A stays on 3, S cannot be on the same floor. So, $S = \{1, 2\}$

Again, R will not allow anyone to stay on 2. SO, $S = 1$.

Rubric:

- 1 point for each correct value (except A).
- 1 point for each explanation.

b) Assume that we have a tree-structured CSP with 100 variables. Each variable can take 5 values. Compare the performances of Tree-CSP-solver with Naïve Backtracking on a computer that can process 10 million nodes/second.

4
(CO1)
(PO1)

Solution: Worst-case solution for naïve backtracking: $n^d = 5^{100}$ nodes.

Time required: $\frac{5^{100}}{10^6} = 2.5 \times 10^{56}$ years.

Worst-case solution for Tree-CSP-solver: $nd^2 = 100 \times 5^2 = 2500$ nodes.

Time required: $\frac{2500}{10^6} = 2.5$ milliseconds.

Rubric:

- 1 point for correct equation.

- 1 point for result.

3. a) “Artificial Intelligence (AI) is the science of making machines that act rationally” - Justify the statement by comparing it with other schools of thought on AI.

7
(CO2)
(PO2)

Solution: Thinking rationally requires finding the rules that govern the correct thought and automating them using big computations. But it does not scale well. Thinking like people requires understanding how brains work and emulating that. But brains are difficult to understand and might not always make the best choice. Acting like people correlates scenarios with human actions to emulate how humans act. But emulating humans does not necessarily mean intelligence as imitating other animals might not give us the best result. Acting rationally focuses on what decisions are made, not the thought process. Each outcome has utilities assigned to them based on our preference. Machines maximize the expected utility.

Rubric:

- 0.5 points for describing each schools of thought.
- 1 point for cons of 3 schools.
- 2 points for pros of acting rationally.

- b) In the state space graph shown in Figure 1, the values beside each arc denote the cost of executing the action for going to a successor state.

3 × 6
(CO1)
(PO1)

Stack:
T6 done
F5
H3
C4
E2
B
A
S1

visit. : SEHCFT
output: SEHCFT

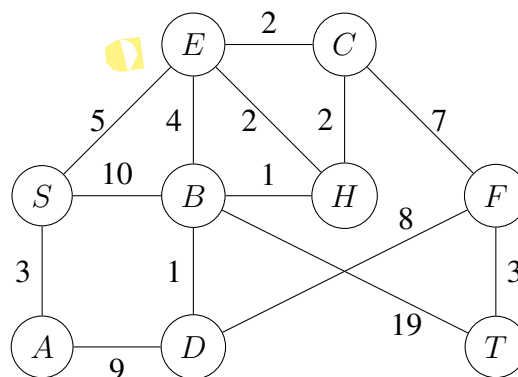


Figure 1: State Space Graph for Question 3(b).

Assume that S is the start state and T is the goal state. We will run graph search variants of different search algorithms to find a path from the S to T . When expanding the successors of a state, we will break ties in alphabetic order.

For each of the following search algorithms, determine the order in which the states are expanded, and the path returned:

- i. Depth-First Search

Solution: States expanded: $S \rightarrow E \rightarrow H \rightarrow C \rightarrow F \rightarrow T$

Path returned: $S \rightarrow E \rightarrow H \rightarrow C \rightarrow F \rightarrow T$

Rubric:

- 3 points for expansion order

- 3 points for the path.

ii. Breadth-First Search

Solution: States expanded: $S \rightarrow A \rightarrow B \rightarrow E \rightarrow D \rightarrow H \rightarrow T$

Path returned: $S \rightarrow B \rightarrow T$

Rubric:

- 3 points for expansion order
- 3 points for the path.

visit: S,S->ABE,A->D,B->HT
sol: SBT

iii. Uniform Cost Search

Solution: States expanded: $S \rightarrow A \rightarrow E \rightarrow C \rightarrow H \rightarrow B \rightarrow D \rightarrow F \rightarrow T$

Path returned: $S \rightarrow E \rightarrow C \rightarrow F \rightarrow T$

Rubric:

- 3 points for expansion order
- 3 points for the path.

S0
S0->A3
S0->E5
E5->C7
E5->H7
S0->B10
A3->D12
C7->F14
F14->T17
T17->T17

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2021-2022
FULL MARKS: 150

CSE 4711: Artificial Intelligence

Answer **all 6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets. The symbols have their usual meanings.

1. Consider that a badly-designed website called 'BeFake' uses a simple encryption software to encrypt the passwords of its users. During registration, BeFake takes the password string as input, encrypts it using a hash function provided by the software, and stores it in a file. If any registered user tries to log in, their input password is again encrypted and checked against the stored password. Unfortunately, you use the same username and password on multiple sites, including BeFake.

Mallory, a malicious hacker and your archenemy, hacked into the server of BeFake and got hold of the encryption software and the files containing the encrypted passwords. She knows your username but does not know the password. However, she knows that your password consists of a combination involving at most three different English letters, 'P', 'Q', and 'R'. She also knows that some letters are more likely to occur than others. So she encodes this by setting, $cost(P) = 1$, $cost(Q) = 2$, and $cost(R) = 3$.

To find your password, Mallory will generate different strings consisting of those letters, provide it to the software to generate the encrypted string, and compare it with the stored password until the first match. She will use search algorithms, specifically, Breadth-First Search (BFS), Depth-First Search (DFS), and Uniform Cost Search (UCS), to generate the passwords in lexicographic order. Assume that all ties are broken in lexicographic order.

- a) Show how Mallory would formulate the search problem by identifying the states, start state, successor function, and goal test.

5
(CO2)
(PO2)

Solution:

State: Each possible string containing the letters, P, Q, and R.

Start State: An empty string.

Successor Function: Appends one letter (P, Q, or R) to the string.

Goal Test: Verify the candidate string using the encryption software.

Rubric:

- 2 points for state
- 1 point each for start state, successor, and goal test.

- b) Assume that the search algorithms, used by Mallory, predict up to 6 letter combinations. Due to collision, the encryption software generates the same hash value for the strings: PPPRR, RQQPPP, RRR, and QQQQ.

20
(CO3)
(PO2)

Assume that your password is RQQPPP. With a brief explanation, assess how doomed you are.

Solution:

BFS will return the shortest possible password, which is RRR. There is no other string shorter than that.

DFS will return the lexicographically smallest password, which is PPPRR.

UCS will return QQQQ since it has the lowest cost (8) among all the strings.

The passwords returned by the algorithms do not match the actual password. However, since the hash values match, Mallory will be able to log in to the badly-designed website using my credentials. The same will happen for all the websites that use the same hash function to encrypt strings. If not, then I am safe (for now).

Rubric:

- 2 points for the correct choice.
- 4 points for the explanation.
- 2 points for the final decision.

2. The St. Petersburg paradox, proposed by Nicolas Bernoulli in 1713, describes the following game: a fair coin is tossed repeatedly by the player until it comes up as tails. If the first tails appears on the n^{th} toss, the player wins 2^n dollars.

a) Given infinite resources, show that the expected monetary value of the lottery is infinite.

6
(CO1)
(PO1)

Solution:

The probability that the first tails appears on the n^{th} toss is $\frac{1}{2^n}$. So,

$$\begin{aligned} EMV(L) &= \sum_{n=1}^{\infty} \frac{1}{2^n} \times 2^n \\ &= \sum_{n=1}^{\infty} 1 \\ &= \infty \end{aligned}$$

Rubric:

- 2 points for probability.
- 2 points for EMV per game.
- 2 points for the conclusion.

b) Assume that Elon Musk (whose net worth is 182.6 billion US Dollars) offers to cover the lottery payout. Show how that affects the expected monetary value of the lottery.

6
(CO3)
(PO2)

Solution:

Since Elon Musk has finite resources, the game must end once those resources are exhausted. Then the maximum number of times a coin can be tossed before it no longer can fully be covered by Mr. Musk is $L = \lfloor \log_2(W) \rfloor$.

Then $EMV(L) = \sum_{n=1}^L \frac{1}{2^n} = \frac{1}{2^n} \times 2^n = L$. So the expected value of the game is:

$$\lfloor \log_2(1.826 \times 10^{11}) \rfloor = 37$$

. So the expected value of the game is \$37.

Rubric:

- 2 points for identifying the problem.

million->6 zeros
billion->9 zeros

- 2 points for determining the equation to calculate the EMV of the lottery.
- 2 points for the actual value.

c) Nicolas's cousin, Daniel Bernoulli, resolved the paradox in 1738 by suggesting that the utility of money is measured on a logarithmic scale. For example, $U(S_m) = \log_2(m)$, where S_m is the state of having \$ m and $m > 0$.

8
(CO3)
(PO2)

Under these circumstances, determine the maximum amount of money a rational player would pay to play the game. Assume that the player has to give up all of his/her money. Use $\sum_{n=1}^{\infty} \frac{n}{2^n} = 2$, if necessary.

Solution:

Assume that the player has \$ k and has to give \$ c to play the game:

$$\begin{aligned}
 U(L) &= \sum_{n=1}^{\infty} \frac{1}{2^n} \times \log_2(k - c + 2^n) \\
 &= \sum_{n=1}^{\infty} \frac{1}{2^n} \times \log_2(2^n) \text{ since } k - c = 0 \\
 &= \sum_{n=1}^{\infty} \frac{n}{2^n} \\
 &= 2
 \end{aligned}$$

To pay the maximum amount to play the game, the player has to have the utility of the given money equal to the utility of the lottery:

$$\begin{aligned}
 \log_2(c) &= 2 \\
 c &= 4
 \end{aligned}$$

So, a rational player would pay \$4 to play the lottery.

Rubric:

- 5 points for the utility of the lottery.
- 2 points for identifying the indifference condition.
- 1 points for the correct amount.

3. The game tree for a game between Alice and Bob is shown in Figure 1. The players make choices to maximize their utility functions which are known to the players beforehand. Alice can move Left, Right, or Straight, whereas Bob can move Left or Right. Alice moves first.

Let (x, y) denote the pair of values in the terminal nodes.

- a) Let the utility functions for Alice and Bob be $U_A(x, y) = x$ and $U_B(x, y) = x - y$ respectively.
- i. Determine the optimal action for Alice in this scenario.

5
(CO3)
(PO2)

Solution:

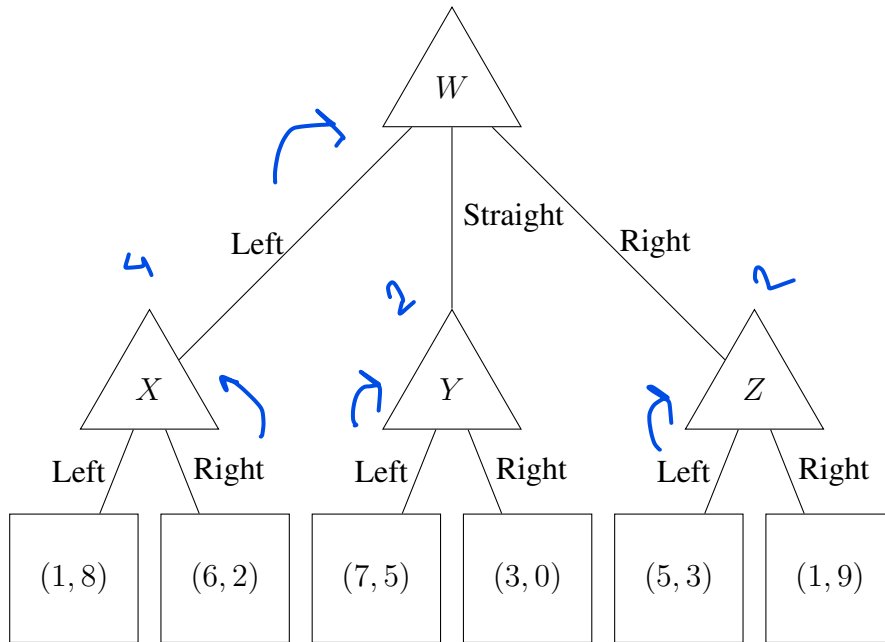


Figure 1: Game Tree for Question 3

$$X = (6, 2)$$

$$Y = (3, 0)$$

$$Z = (5, 3)$$

$$W = (6, 2)$$

Since X gives Alice the maximum utility, she should move Left.

Rubric:

- 1 point for each value.
- 1 point for correct action.

- ii. Identify the nodes that can be pruned considering the branches are explored from left to right. 1
(CO1)
(PO1)

Solution:

None.

Rubric:

- 2 points for the correct answer.

- b) Assume that Bob's utility function changes based on his mood. Table 1 denotes the marginal probability distribution associated with Bob's mood and Table 2 denotes the conditional probability distribution associated with Bob's utility function conditioned on his mood.

Table 1: Mood distribution

M	P
Happy	0.60
Sad	0.40

Table 2: Utility distribution

U_B	$P(U_B M = \text{Happy})$	$P(U_B M = \text{Sad})$
$-x$	0.45	0.30
$x - y$	0.25	0.45
$x^2 + y^2$	0.30	0.25

i. Calculate the likelihood of each utility functions for Bob.

3 × 2
(CO3)
(PO2)

Solution:

$$\begin{aligned} P(U_b(x, y) = -x) &= P(M = \text{Happy}) \times P(U_B = -x | M = \text{Happy}) \\ &\quad + P(M = \text{Sad}) \times P(U_B = -x | M = \text{Sad}) \\ &= 0.60 \times 0.45 + 0.40 \times 0.30 \\ &= 0.39 \end{aligned}$$

$$\begin{aligned} P(U_b(x, y) = x - y) &= P(M = \text{Happy}) \times P(U_B = x - y | M = \text{Happy}) \\ &\quad + P(M = \text{Sad}) \times P(U_B = x - y | M = \text{Sad}) \\ &= 0.60 \times 0.25 + 0.40 \times 0.45 \\ &= 0.33 \end{aligned}$$

$$\begin{aligned} P(U_b(x, y) = x^2 + y^2) &= P(M = \text{Happy}) \times P(U_B = x^2 + y^2 | M = \text{Happy}) \\ &\quad + P(M = \text{Sad}) \times P(U_B = x - y | M = \text{Sad}) \\ &= 0.60 \times 0.30 + 0.40 \times 0.25 \\ &= 0.28 \end{aligned}$$

Rubric:

- 1 point for each correct equation.
- 1 point for each correct value.

ii. Determine the optimal action for Alice in this scenario.

13
(CO3)
(PO2)

Solution:

From X, Bob will choose:

- (1, 8) when $U_B(x, y) = -x$, resulting in Alice getting 1.
- (6, 2) when $U_B(x, y) = x - y$, resulting in Alice getting 6.
- (1, 8) when $U_B(x, y) = x^2 + y^2$, resulting in Alice getting 1.

Expected utility of X for Alice = $0.39 \times 1 + 0.33 \times 6 + 0.28 \times 1 = 2.65$. Sum of $P(x) \cdot U(x)$

From Y, Bob will choose:

- (3, 0) when $U_B(x, y) = -x$, resulting in Alice getting 3.
- (3, 0) when $U_B(x, y) = x - y$, resulting in Alice getting 3.
- (7, 5) when $U_B(x, y) = x^2 + y^2$, resulting in Alice getting 7.

Expected utility of Y for Alice = $0.39 \times 3 + 0.33 \times 3 + 0.28 \times 7 = 4.12$.

From Z, Bob will choose:

- (1, 9) when $U_B(x, y) = -x$, resulting in Alice getting 1.
- (5, 3) when $U_B(x, y) = x - y$, resulting in Alice getting 5.
- (1, 9) when $U_B(x, y) = x^2 + y^2$, resulting in Alice getting 1.

Expected utility of Z for Alice = $0.39 \times 1 + 0.33 \times 5 + 0.28 \times 1 = 2.32$.
 Since Y gives Alice the maximum expected utility, she should move Straight.

Rubric:

- 1 point for each correct choice for Bob and Alice.
- 1 point for each expected utility.
- 1 points for correct action.

4. Consider that a game of Pacman is being played on the grid shown in Figure 2.

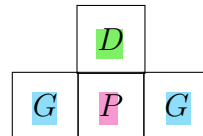


Figure 2: Pacman Grid for Question 4

Here, P indicates the position of Pacman, G indicates the position of the ghosts, and D indicates the position of a food dot. For simplicity, the ghosts remain stationary throughout the game.

To determine the policy for playing the game, our feature-based Q-learning agent, Pacman, uses two features, F_g and F_d defined as follows:

$$F_g(s, a) = F_1(s) + F_2(s, a) + F_3(s, a)$$

$$F_d(s, a) = F_4(s) + 2 \times F_5(s, a)$$

where

$F_1(s)$ = number of ghosts within 1 step of state s

$F_2(s, a)$ = number of ghosts Pacman touches after taking action a from state s

$F_3(s, a)$ = number of ghosts within 1 step of the state Pacman ends up in after taking action a

$F_4(s)$ = number of food dots within 1 step of state s

$F_5(s, a)$ = number of food dots eaten after taking action a from state s

After a few episodes of Q-learning, the weights are $w_g = -10$ and $w_d = 100$. The discount factor, $\gamma = 0.5$ and learning rate, $\alpha = 0.5$. The action space of Pacman is $\{left, right, up, down, stay\}$. Pacman can take any actions from a state given it does not go beyond the grid.

a) Considering the actions, a that are available from the current position, s of Pacman:

i. Calculate $F_g(s, a)$ and $F_d(s, a)$.

12+6
(CO1)
(PO1)

Solution:

$$\begin{aligned}
 F_g(s, left) &= 2 + 1 + 1 = 4 \\
 F_g(s, right) &= 2 + 1 + 1 = 4 \\
 F_g(s, up) &= 2 + 0 + 0 = 2 \\
 F_g(s, stay) &= 2 + 0 + 2 = 4 \\
 F_d(s, left) &= 1 + 2 \times 0 = 1 \\
 F_d(s, right) &= 1 + 2 \times 0 = 1 \\
 F_d(s, up) &= 1 + 2 \times 1 = 3 \\
 F_d(s, stay) &= 1 + 2 \times 0 = 1
 \end{aligned}$$

Rubric:

- 0.5 point for each correct equation.
- 1 point for each correct value.

ii. Calculate $\hat{Q}(s, a)$.

Solution:

$$\hat{Q}(s, up) = w_g F_g(s, up) + w_d F_d(s, up) = (-10) \times 2 + 100 \times 3 = 280$$

$$\hat{Q}(s, left) = w_g F_g(s, left) + w_d F_d(s, left) = (-10) \times 4 + 100 \times 1 = 60$$

$$\hat{Q}(s, right) = w_g F_g(s, right) + w_d F_d(s, right) = (-10) \times 4 + 100 \times 1 = 60$$

$$\hat{Q}(s, stay) = w_g F_g(s, stay) + w_d F_d(s, stay) = (-10) \times 4 + 100 \times 1 = 60$$

Rubric:

- 0.5 points for each correct equation.
- 1 point for each Q-Value.

b) Determine the optimal policy for Pacman from its current position following the $\hat{Q}(s, a)$ values calculated. Argue on how good the policy is considering the alternatives.

2 + 3
(CO3)
(PO2)

Solution:

The optimal policy for the current position of Pacman should be:

$$\pi(s) = \operatorname{argmax} (\hat{Q}(s, up), \hat{Q}(s, left), \hat{Q}(s, right), \hat{Q}(s, stay)) \\ = up$$

Since going up results in Pacman eating the food dot, it is reasonable to decide to go up. Compared to other moves, where Pacman stays in the same position or gets eaten by the ghost, this is the best possible outcome for Pacman.

Rubric:

- 2 points for the correct decision.
- 3 points for the argument.

c) From its current position, s , Pacman moves up to go to the cell, s' containing the food dot and eats it. We observe a reward, $R(s, a, s') = 250$.

4 + 3
(CO1)
(PO1)

Considering the actions, a' that are available from s' :

i. Calculate $Q(s, up)$.

Solution:

$$\begin{aligned} Q(s, up) &= R(s, a, s') + \gamma \times \max_{a'} \hat{Q}(s', a') \\ &= 250 + 0.5 \times \max (\hat{Q}(s', down), \hat{Q}(s', stay)) \\ &= 250 + 0.5 \times 0 \\ &= 250 \end{aligned}$$

it can only move up if prev move was either "stay" or "down"

where

$$\hat{Q}(s', \text{down}) = w_g F_g(s', \text{down}) + w_d F_d(s', \text{down}) = (-10) \times 2 + 100 \times 0 = -20$$

$$\hat{Q}(s', \text{stay}) = w_g F_g(s', \text{stay}) + w_d F_d(s', \text{stay}) = (-10) \times 0 + 100 \times 0 = 0$$

Rubric:

- 1 point for the correct equation.
- 1 point for each of the correct $Q(s', a')$.
- 1 point for the final result.

ii. Update w_g and w_d . since we are going up

Solution:

$$w_g = w_g + \alpha \times (Q(s, \text{up}) - \hat{Q}(s, \text{up})) \times F_g(s, a) = -10 + 0.5 \times (250 - 280) \times 2 = -40$$

$$w_d = w_d + \alpha \times (Q(s, \text{up}) - \hat{Q}(s, \text{up})) \times F_d(s, a) = 100 + 0.5 \times (250 - 280) \times 3 = 55$$

Rubric:

- 0.5 points for each equation.
- 1 point for each correct value.

5. a) Consider the following statements:

- Surely computers cannot be intelligent - they can do only what their programmers tell them.
- Surely humans cannot be intelligent - they can do only what their genes tell them.

12
(CO3)
(PO2)

For each of the statements, is the latter clause true? And does it imply the former? Provide brief arguments.

Solution:

This depends on the definition of “intelligent” and “tell”. In one sense computers/humans only do what their programmers/genes command them to do, but in another sense what the programmers/genes tells the computer/human to do often has very little to do with what the computer/human actually does.

For a computer, in one sense, a programmer can “tell” the computer “learn to play chess better than I do, and then play that way”. But in another sense, the programmer told the computer “follow this learning algorithm” and it learned to play. So we are left in the situation where we may or may not consider learning to play chess to be a sign of intelligence, and we may think the intelligence resides in the programmer or in the computer.

The second statement is parallel with the first one. Whatever we decide about whether computers could be intelligent, we are committed to making the same conclusion about humans, unless our reasons for deciding whether something is intelligent take into account the mechanism.

Rubric: This question is intended to stimulate discussion. So there is no perfect answer. However, the argument of the student, based on what they have learnt throughout the semester, should discuss the following:

- The role of programmers/genes in AI

- The role of the behavior of computers/humans
- The parallel between these two
- Concluding remarks on their decision

3 points for each part.

- b) Sometimes Markov Decision Processes (MDP) are formulated with a reward function $R(s, a)$ that depends on the action taken, or $R(s, a, s')$ that also depends on the outcome state. Modify the Bellman Equation to determine the value of a state based on these formulations.

5
(CO2)
(PO2)

Solution:

For $R(s, a)$, we have:

$$V(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} T(s, a, s') V(s') \right]$$

And for $R(s, a, s')$, we have:

$$V(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V(s')]$$

Rubric:

- 2.5 points for each formulation.

- c) Suppose that we define the value of a state to be the **maximum** (as opposed to summation) reward obtained from its future states. Does this result in stationary preferences? If yes, justify your position. If not, provide a counterexample.

2 + 6
(CO1)
(PO1)

Solution:

Stationary preferences require the agent to follow:

$$[a_1, a_2, \dots] \succ [b_1, b_2, \dots]$$



$$[r, a_1, a_2, \dots] \succ [r, b_1, b_2, \dots]$$

However, if the utility of a sequence is its **maximum reward**, we can easily violate the stationary preferences. For example,

$$[4, 3, 0, 0, \dots] \sim [4, 0, 0, 0, \dots]$$

but if we remove the first item from both sequences:

$$[3, 0, 0, \dots] \succ [0, 0, 0, \dots]$$

So it breaks stationary preferences.

Rubric:

- 2 points for decision.
- 3 points for the theorem.
- 3 points for the counterexample.

6. Consider that we have placed a robot in an unknown environment. It can perceive its current state,

s_t , and execute an action, a_t . Upon executing the action, the robot can move to another state, s_{t+1} (possibly the same as before), and get an instant reward, r_t . The robot moves around for a while and comes up with the experiences shown in Table 3.

Table 3: Experience achieved in 5 iterations for Question 6

t	s_t	a_t	s_{t+1}	r_t
0	A	Up	B	2
1	B	Up	B	-4
2	B	Down	B	0
3	B	Down	A	3
4	A	Down	A	-1

No transition function given, but experience given

Now, answer the following questions based on the scenario mentioned above:

- a) Formulate the scenario as a sample-based Q-learning problem by determining Q-values for each state-action pair based on the experience of the robot.

22

(CO2)

(PO2)

Assume that the discount factor, $\gamma = 0.5$ and the learning rate, $\alpha = 0.5$.

Solution:

At $t = 0$, we get: initially all $Q = 0$

$$\begin{aligned} Q(A, Up) &= (1 - \alpha)Q(A, Up) + \alpha(r + \gamma \max_{a'} Q(B, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (2 + 0.5 \times 0) \\ &= 1 \end{aligned}$$

At $t = 1$, we get:

$$\begin{aligned} Q(B, Up) &= (1 - \alpha)Q(B, Up) + \alpha(r + \gamma \max_{a'} Q(B, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (-4 + 0.5 \times 0) \\ &= -2 \end{aligned}$$

At $t = 2$, we get:

$$\begin{aligned} Q(B, Down) &= (1 - \alpha)Q(B, Down) + \alpha(r + \gamma \max_{a'} Q(B, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (0 + 0.5 \times 0) \\ &= 0 \end{aligned}$$

At $t = 3$, we get:

$$\begin{aligned} Q(B, Down) &= (1 - \alpha)Q(B, Down) + \alpha(r + \gamma \max_{a'} Q(A, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (3 + 0.5 \times 1) \\ &= 1.75 \end{aligned}$$

At $t = 4$, we get:

$$\begin{aligned} Q(A, Down) &= (1 - \alpha)Q(A, Down) + \alpha(r + \gamma \max_{a'} Q(A, a')) \\ &= (1 - 0.5) \times 0 + 0.5 \times (-1 + 0.5 \times 1) \\ &= -0.25 \end{aligned}$$

So the final values are:

$$Q(A, Down) = -0.25$$

$$Q(A, Up) = 1$$

$$Q(B, Down) = 1.75$$

$$Q(B, Up) = -2$$

Rubric:

- 4 points for each iteration.
- 0.5 points for each value.

b) Determine the optimal policy from the scenario.

3
(CO3)
(PO2)

Solution:

$$\begin{aligned}\pi(A) &= \operatorname{argmax}(Q(A, down), Q(A, up)) \\ &= Up\end{aligned}$$

$$\begin{aligned}\pi(B) &= \operatorname{argmax}(Q(B, down), Q(B, up)) \\ &= Down\end{aligned}$$

Rubric:

- 1.5 points for each correct policy.

CSE 4711 - Quiz #4

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CSE 4711 - Quiz #4

Which of the following DOES NOT require approximating transition function to perform its updates?

- ☒ Approximate Q-Learning
- ☒ Temporal Difference Learning
- ☒ Direct Evaluation
- ☒ Q-Learning

Which of the following requires considering exploration vs. exploitation tradeoff?

- ☐ Direct Evaluation
- ☐ Temporal Difference Learning
- ☒ Q-Learning
- ☐ Approximate Q-Learning



Having a highly negative living reward discourages exploration.

☒ True

☐ False

Clear selection

If we want to minimize regret, which of the following exploration strategy should we choose?

☐ Epsilon-greedy

☒ Exploration function

Clear selection

Reinforcement learning is similar to Markov Decision Process (MDP), except for the fact that we work with existing policies, instead of starting with a blank one.

☐ True

☒ False

Clear selection

Given enough memory to store the Q-values of the states, we can use Q-learning to determine the optimal policy in polynomial time.

☐ True

☒ False

Clear selection



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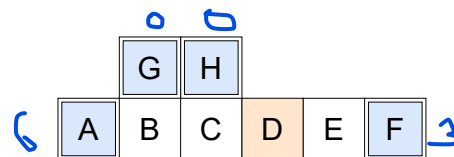
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QUIZ 3**DURATION: 30 Minutes****WINTER SEMESTER, 2020-2021****FULL MARKS: 30****CSE 4711: Artificial Intelligence**There are **2 (two)** questions. Answer **both** of them.

Figures in the right margin indicate marks for each question.

The square brackets at the start of each question denote the corresponding CO and PO.

1. Consider that an agent is in cell **D** of the Gridworld shown in Figure 1. The only action available in cells **A, F, G, and H** is **Exit** with rewards 6, 3, 0, and 0, respectively. When the agent exits from a cell, it gets the reward for that cell as specified. From the other cells, the agent can only take the action **Left** or **Right**, which results in the agent moving to the immediate left or right cell, respectively.

**Figure 1:** Gridworld for Question 1

If the agent is in cell **B** or **C**, and it takes a **Left** or **Right** action, it might fail sometimes. The agent will move in the desired direction with probability p , and it will fail and move up with probability $1 - p$. If the agent is in any other cell (**A, D, E, F, G, or H**), the action will always be successful.

Assume that there is **no living reward/penalty** and the **discount factor is γ** .

- a) [CO2, PO4] Model the problem as an MDP. Show the set of states, set of actions, start state, and terminal state(s). 8

Solution:

- States, $S = \{A, B, C, D, E, F, G, H\}$
- Actions, $A = \{\text{Left, Right, Exit}\}$
- Start State: **D**
- Terminal States: $\{A, F, G, H\}$

Rubric:

- 0.5 points for each correct value

- b) [CO1, PO1] For each of the following policies, determine the value of each non-terminal state. Assume $\gamma \in [0, 1]$. 6 × 2

i. $\pi_{\text{Right}}(S) = \begin{cases} \text{Exit}, & \text{if } S \text{ is a Terminal State} \\ \text{Right}, & \text{otherwise} \end{cases}$

Solution:

- $B \rightarrow 3\gamma^4 p^2$
- $C \rightarrow 3\gamma^3 p$
- $D \rightarrow 3\gamma^2$
- $E \rightarrow 3\gamma$

(A, D, E, F, G, or H), the action will always be successful.
Here we ignore (1-p) as reward is zero. otherwise write that part also

Rubric:

- 0.5 points for each correct integer
- 0.5 points for each correct discount
- 0.5 points for each correct probability

ii. $\pi_{\text{Left}}(S) = \begin{cases} \text{Exit}, & \text{if } S \text{ is a Terminal State} \\ \text{Left}, & \text{otherwise} \end{cases}$

Solution:

- $B \rightarrow 6\gamma p$
- $C \rightarrow 6\gamma^2 p^2$
- $D \rightarrow 6\gamma^3 p^2$
- $E \rightarrow 6\gamma^4 p^2$

(A, D, E, F, G, or H), the action will always be successful.

Rubric:

- 0.5 points for each correct integer
- 0.5 points for each correct discount
- 0.5 points for each correct probability

c) [CO3, PO2, PO3] What values of p will ensure that going Right from the start state is the optimal action? Provide a range in terms of γ .

5

Solution: For going right to be optimal, we need $V^{\pi_{\text{Right}}}$ of the start state to be greater or equal to $V^{\pi_{\text{Left}}}$. We get:

$$V^{\pi_{\text{Right}}}(D) \geq V^{\pi_{\text{Left}}}(D)$$

$$3\gamma^2 \geq 6\gamma^3 p^2$$

$$6\gamma^3 p^2 \leq 3\gamma^2$$

$$p^2 \leq \frac{3}{6\gamma}$$

$$p \leq \frac{1}{\sqrt{2\gamma}}$$

We get: $p \in \left[0, \frac{1}{\sqrt{2\gamma}}\right]$

Rubric:

- For defining the correct relation between the value of the start state: 3 points
- For the correct lower range: 1 point
- For the correct upper range: 1 point

2. [CO1, CO2, CO3, PO1, PO2, PO3, PO4] Write everything that you know about MDPs that was not asked in this quiz!

5

Solution: Rubric:

- Any correct information provided here that is not related to the questions given are accepted.

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QUIZ 2**DURATION: 30 Minutes****WINTER SEMESTER, 2020-2021****FULL MARKS: 30****CSE 4711: Artificial Intelligence**There are **2 (two)** questions. Answer **both** of them.

Figures in the right margin indicate marks.

1. Consider that you are working for *Bingo*, a lottery insurance company that offers people flat payments in exchange for their lottery tickets. There's a lottery where people have $p\%$ chance of winning $\$M$ and $(100 - p)\%$ chance of winning $\$0$. You are designing a system to determine the payment rate for people to persuade them to give up their lottery tickets. Using data mining techniques, you have determined their utility functions, $U(\$X)$, where X is the amount of money. Your system will be given the p , $U(\$X)$, and M for each person. It will output the minimum payment required to convince them to give up the lottery ticket. Assume that people are rational agents.

- a) [CO2, PO4] Model the system based on the given requirements.

10

Solution: Let, L be the lottery event. Then,

$$U(L) = \frac{p \times U(\$M)}{100} + \frac{(100 - p) \times U(\$0)}{100}$$

To convince a rational agent to give up the lottery ticket and accept flat payment, we need to pay more than $\$U(L)$.

Rubric:

- For correct probabilities: 4 points
- For assuming that money is not equal to the utility of money: 4 points
- For predicting that we need to give more than the utility of the lottery: 2 point

- b) [CO1, PO1] For each of the following scenarios, predict the amount of flat payment required to make people indifferent to playing the lottery and receiving the flat payment.

3 × 2

- i. $p = 0.5$, $M = 10^8$, $U(\$X) = \sqrt{X}$

Solution: Using the formula from 1(a), we need to pay (in USD):

$$\begin{aligned} U(L) &= \frac{0.5 \times U(10^8)}{100} + \frac{(100 - 0.5) \times U(0)}{100} \\ &= \frac{0.5 \times \sqrt{10^8}}{100} + \frac{99.5 \times \sqrt{0}}{100} \\ &= 50 \end{aligned}$$

Rubric:

- For placing correct values in equation/answer: 2 points
- For predicting that we need to pay money equal to the utility of the lottery: 1 point

ii. $p = 30, M = 10^3, U(\$X) = X + 5$

Solution: Using the formula from 1(a), we need to pay (in USD):

$$\begin{aligned} U(L) &= \frac{30 \times U(10^3)}{100} + \frac{(100 - 30) \times U(0)}{100} \\ &= \frac{30 \times (10^3 + 5)}{100} + \frac{70 \times (0 + 5)}{100} \\ &= 305 \end{aligned}$$

Rubric:

- For placing correct values in equation/answer: 2 points
- For predicting that we need to pay money equal to the utility of the lottery: 1 point

2. Consider that a Pacman game is being played in the map shown below:

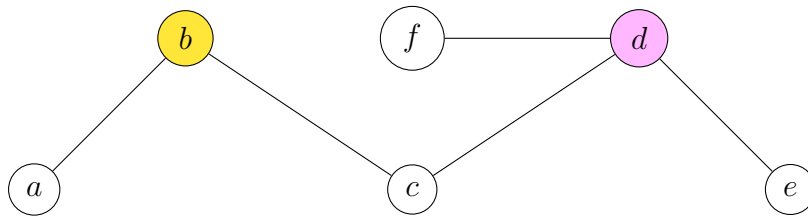
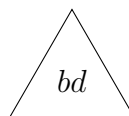


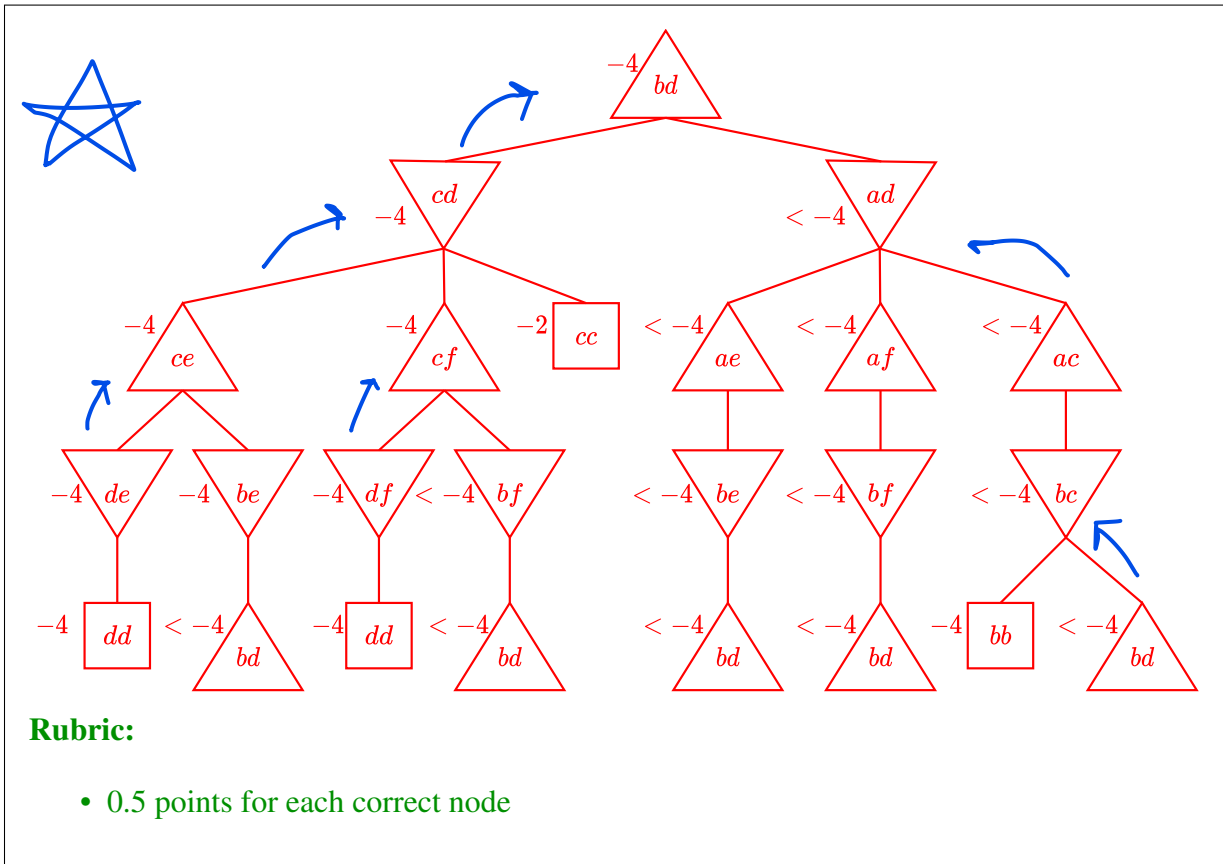
Figure 1: Pacman Game Map for Question 2

Initially, Pacman is in node b and the ghost is in node d . The ghost wants to catch Pacman, and Pacman wants to evade it. Pacman and the ghost will take turns moving from one node to another and Pacman moves first. In their turn, the player (Pacman/Ghost) must move to an adjacent node. The game ends when the ghost catches Pacman. Each edge requires 1 second to cross. The terminal payoff of the Ghost is negative of the total time taken.

- a) [CO2, PO4] Draw the Minimax Tree up to depth 4 (maximum two moves for Pacman, and maximum two moves for Ghost). Use up triangles ($\triangle$) for maximizer nodes, down triangles (∇) for minimizer nodes, and squares ($\square$) for terminal nodes. Denote each node in the Minimax Tree as pair of values, where the first value denotes the position of Pacman, and the second value denotes the position of the ghost at that state. For example, the root node will look like this:



Solution:



b) [CO1, PO1] What is the maximum score the ghost can get if both play optimally? Provide a brief explanation.

2

Solution: From the top value of the Minimax Tree, the answer is -4.

Alternate solution: From the graph, it can be seen that the best the ghost can do is go towards Pacman. An optimal Pacman would try to run away from the ghost, but eventually it will be caught. It requires 4 steps to catch Pacman.

Rubric:

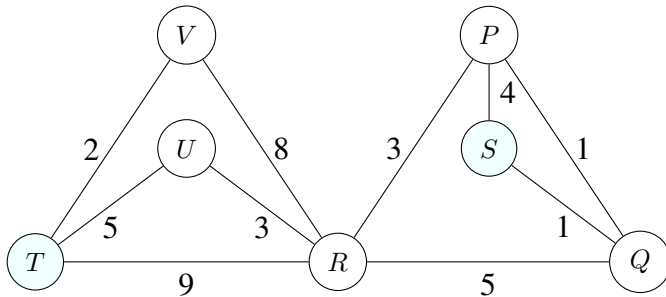
- 1 point for the correct answer
- 1 point for correct explanation.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 1**DURATION: 30 Minutes****WINTER SEMESTER, 2020-2021****FULL MARKS: 30****CSE 4711: Artificial Intelligence**There are **2 (two)** questions. Answer **both** of them.

Figures in the right margin indicate marks.

Consider the following state space graph and the table containing two heuristic functions:



Node	h_0	h_1
P	8	10
Q	9	12
R	7	8
S	9.5	10
T	0	0
U	4	4.5
V	1.5	1

Figure 1: State Space Graph for Question 1**Table 1:** Heuristic Functions for Question 1

In Figure 1, **S** is the start state and **T** is the goal state. The cost for taking each edge (in both directions) is shown on the graph. In Table 1, h_0 is consistent, but h_1 is not consistent.

1. We can apply *Graph Search Strategy* to find a path from start state to the goal state using:

 5×3

- Depth First Search
- Breadth First Search
- Uniform Cost Search
- A* Search with Heuristic h_0
- A* Search with Heuristic h_1

Which of the following paths will be returned for each of the algorithms mentioned above?

- (i) $S \rightarrow P \rightarrow R \rightarrow T$
- (ii) $S \rightarrow Q \rightarrow R \rightarrow T$
- (iii) $S \rightarrow Q \rightarrow P \rightarrow R \rightarrow U \rightarrow T$

Understand that for some of the algorithms mentioned above, the path from the start state to the goal state can vary based on how we break ties among multiple options. In such cases, mention *all* possible paths returned.

Solution:

- DFS: (i), (ii), (iii)
- BFS: (i), (ii)
- UCS: (iii)
- A* with h_0 : (iii)
- A* with h_1 : (iii)

Rubric:

- For each of the correctly identified paths: 1 point
- For each of the correctly omitted paths: 1 point

2. You are designing a new heuristic function h_2 for the given state space graph in Figure 1. You have come up with all the values except $h_2(Q)$.

Node	P	Q	R	S	T	U	V
h_2	9	?	7	10	0	4.5	1.5

For each of the following scenarios, write the possible range of values for $h_2(Q)$ with a brief explanation:

a) What values $h_2(Q)$ will make h_2 admissible?

4

Solution: To make h_2 admissible, $h_2(Q)$ has to be less than or equal to the actual optimal cost from Q to goal T , which is the cost of the path $Q \rightarrow P \rightarrow R \rightarrow U \rightarrow T$, i.e. 12. And the minimum possible value for heuristic is no heuristic, which makes all the values 0. The answer is $0 \leq h_2(Q) \leq 12$.

Rubric:

- For the correct range: 2 points
- For a correct explanation of the lower bound: 1 point
- For a correct explanation of the upper bound: 1 point

b) What values of $h_2(Q)$ will make h_2 consistent?

5

Solution: All the other nodes except node Q satisfy the consistency conditions. The consistency conditions that involve the state Q are:

$$\begin{aligned} \bullet \quad h(S) &\leq c(S, Q) + h(Q) \\ &\rightarrow 10 \leq 1 + h(Q) \\ &\rightarrow 9 \leq h(Q) \end{aligned}$$

$$\begin{aligned} \bullet \quad h(Q) &\leq c(Q, S) + h(S) \\ &\rightarrow h(Q) \leq 1 + 10 \\ &\rightarrow h(Q) \leq 11 \end{aligned}$$

$$\begin{aligned} \bullet \quad h(P) &\leq c(P, Q) + h(Q) \\ &\rightarrow 9 \leq 1 + h(Q) \\ &\rightarrow 8 \leq h(Q) \end{aligned}$$

$$\begin{aligned} \bullet \quad h(Q) &\leq c(Q, P) + h(P) \\ &\rightarrow h(Q) \leq 1 + 9 \\ &\rightarrow h(Q) \leq 10 \end{aligned}$$

$$\begin{aligned} \bullet \quad h(R) &\leq c(R, Q) + h(Q) \\ &\rightarrow 7 \leq 5 + h(Q) \\ &\rightarrow 2 \leq h(Q) \end{aligned}$$

$$\begin{aligned} \bullet \quad h(Q) &\leq c(Q, R) + h(R) \\ &\rightarrow h(Q) \leq 5 + 7 \\ &\rightarrow h(Q) \leq 12 \end{aligned}$$

All connected goes through Q

Q goes through all connected

Using the tightest bounds from the equations above: $9 \leq h_2(Q) \leq 10$

Rubric:

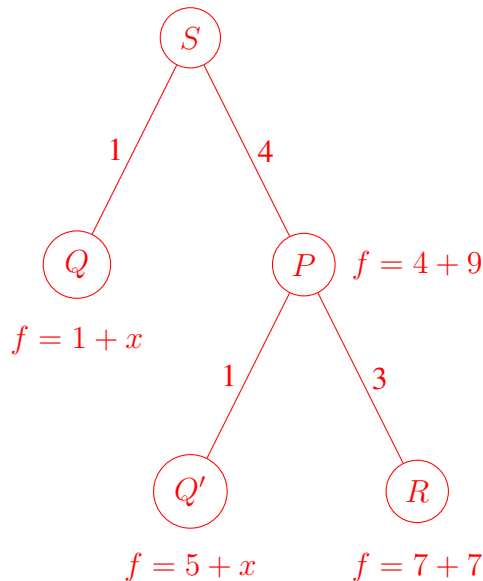
- For the correct range: 2 points

- For a correct explanation of the lower bound: 1.5 point
- For a correct explanation of the upper bound: 1.5 point

c) What values of $h_2(Q)$ will cause A* Graph Search to expand node S , then node P , then node Q , then node R in order?

6

Solution: Let, $h_2(Q) = x$.



The A* search tree using the heuristic h_2 is shown above. Node S is expanded first since it's the start state.

To make A* graph search expand node P before node Q , we need

$$1 + x > 13$$

$$x > 12$$

After expanding P , we need to either expand Q' or Q before R . To expand Q' ,

$$5 + x < 14$$

$$x < 9$$



which violates the earlier constraint. To expand Q ,

$$1 + x < 14$$

$$x < 13$$



So we can get $12 < h_2(Q) < 13$

Rubric:

- For the correct range: 2 points
- For a correct explanation of the lower bound: 2 points
- For a correct explanation of the upper bound: 2 points

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION

WINTER SEMESTER, 2020-2021

DURATION: 1 Hour 30 Minutes

FULL MARKS: 75

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed.

There are 3 (three) questions. Answer all 3 (three) of them.

Figures in the right margin indicate marks of each question.

The square brackets on the start of each question denotes the corresponding CO and PO.

1. There are around 4.3×10^{19} possible configurations of a 3×3 Rubik's cube. However, if played optimally, any configuration can be solved in 20 moves or less. Here, one single move consists of a rotation of one of the faces of the cube. There are 27 possible rotations from a single configuration. We can pose the problem of solving Rubik's cube as a search problem. Assume that the closest solution from our start state requires exactly 20 moves.

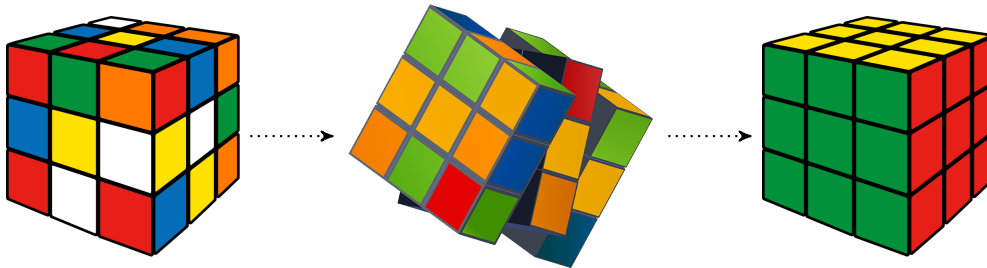


Figure 1: Rubik's Cube

- a) [CO2, PO4] Model the state space graph for this problem.

10

Solution: State: Each possible configuration of the cube corresponds to a state. There are 4.3×10^{19} states.

Actions: Rotations. There are 27 actions available from each state.

Successor Function: Go from one state to another state with a single rotation. There will be an edge connected to the state. The cost can be the number of moves.

Start State: Given the initial configuration of the Rubik's cube.

Goal Test: For each face, the colors of the individual cubelets on that face are the same.

Rubric:

- 2 points for each portion

- b) [CO2, PO4] Model the search tree from the state space graph.

3

Solution: Branching Factor: 27, since we can make 27 moves from each state.

Maximum Depth: 20, since any configuration can be solved in 20 moves or less.

Solution Depth: As mentioned in the question, the depth of the solution is 20.

Rubric: less than or equal to 20

- 1 point for each portion

- c) [CO3, PO2] Considering both the tree search and graph search variants of Breadth-First Search (BFS) and Depth-First Search (DFS), which algorithm will you prefer to find the solution? Justify your choice.

Solution: (This is one of the possible answers. Other logical answers are acceptable as well)

Since the solution is at the last level of the search tree, we should opt for DFS over BFS. Because BFS tree search will explore 27^{20} nodes in both the best and the worst case. BFS graph search performs a bit better, expanding 4.3×10^{19} nodes in both the best and the worst case. On the other hand, the DFS tree search can get stuck in a loop since the state space graph can contain cycles. However, the best case solution would require expanding only 20 nodes. The worst-case for DFS graph search occurs if the solution is on the rightmost node of the tree and we go from left to right. In this case, we will expand 4.3×10^{19} nodes. However, the best case can expand only 20 nodes. So the solution would require expanding anything between $[20, 4.3 \times 10^{19}]$ nodes.

Since the state space graph is huge, we can go for a DFS graph search. This will find a solution with possibly less number of nodes expansion than BFS, while also using a relatively small amount of memory.

Rubric:

- Reasoning for the choice: 6 points
- Reasoning for not picking other algorithms: 2×3 points

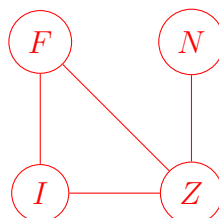
2. Zahid (Z), Ishrak (I), Farabi (F), and Nafisa (N) came to Chini-Come, a restaurant near their university. The restaurant serves Special Rice (S), Biryani Rice (B), Kashmiri Naan (K), and Paratha (P). You overhear their conversations, and come up with the following preferences:

- Zahid does not like Paratha.
- Ishrak and Farabi want to grab a bite of each other's food. So they want to order different dishes.
- Farabi likes Rice items. So he'll either take Special Rice or Biryani Rice.
- Zahid wants to take a unique dish. However, he loves to copy Ishrak and will order the same dish as Ishrak.
- Nafisa will not order Kashmiri Naan as she had them earlier.

Now you want to figure out who will order what using CSP.

- a) [CO2, PO4] Draw the constraint graph considering each person as a variable.

Solution: Constraint Graph:



Rubric:

- 0.5 points for each node.

- 0.5 points for each correctly included edge.
- 0.5 points for each correctly omitted edge.

b) [CO1, PO1] Show the remaining values in the domains of each variable after enforcing the unary constraints.

8

Solution:

- $Z \rightarrow S, B, K$
- $I \rightarrow S, B, K, P$
- $F \rightarrow S, B$
- $N \rightarrow S, B, P$

Rubric:

- 0.5 points for each correctly included value
- 0.5 points for each correctly omitted value

c) [CO1, PO1] Based on the result in Question 2(b), determine the cutset that will result in the least number of residual CSPs and provide brief reasoning for your choice. How many residual CSPs will you get?

2 + 1

Solution: Since F has the least remaining values, removing it will result in 2 residual CSPs. So we'll pick F .

Rubric:

- Correct choice: 1 point
- Explanation: 1 point
- Residual CSP count: 1 point

d) [CO3, PO2] Let's say Ishrak orders Biryani Rice. Based on the result in Question 2(b), perform Arc Consistency and show the remaining values for the domain of each variable. What would have been the difference if we performed Forward Checking here?

8 + 1

Solution:

- $Z \rightarrow B$
- $I \rightarrow B$
- $F \rightarrow S$
- $N \rightarrow S, P$

If we performed forward checking, it would not have considered the $N - Z$ edge, and would not remove S from the domain of N .

Rubric:

- 0.5 points for each correctly included value
- 0.5 points for each correctly omitted value
- 1 point for forward checking explanation

3. Consider that you have an encryption software that takes a string, and converts it to an encrypted string. You used this software to encrypt your password and store it in a file, but somehow you forgot the password. You know that the password contains only the letters X , Y , and Z . The only way to recover the password is to provide a guess to the software, which will give you the encrypted string and then compare it with the string stored in the file. You are trying to recover your password using search.

You formulate your search problem in this manner:

State: Each possible string consisting of the letters X , Y , Z

Start State: An empty string

Successor Function: Appends one letter (X , Y , or Z) to the string

Path Cost: You suspect that some letters are more likely to occur than others. To encode this in your search, you set $cost(X) = 1, cost(Y) = 2, cost(Z) = 3$.

Goal Test: Verify a candidate password using the decryption software.

Your search algorithm will keep generating different passwords and send them to the encryption software for checking until you find the first match. Assume that all ties are broken alphabetically.

- a) [CO3, PO3] Assume that your search algorithm predicts up to 10 character strings. There are 6 correct passwords: $XXXZZZ$, $XYZZZ$, $YXYXY$, $YZXYXZY$, $ZYXZ$, and $ZYXZY$. Matching with any one of them will do. With a brief explanation, select the one that will be returned by: 6×3

- i. Depth-First Search (DFS)

Solution: Depth-first search will expand states in alphabetical order: X , XX , XXX , ..., $XXXXXXXXXX$, $XXXXXXXXXXY$, With alphabetical tie-breaking, the depth-first search will return the correct password that sorts first alphabetically: in this case, $XXXZZZ$.

Rubric:

- 2 points for correct choice
- 4 points for correct explanation

only DFS is not optimal

- ii. Breadth-First Search (BFS)

Solution: Breadth-first search will return the shortest correct password, $ZYXZ$. All of the other passwords are longer, so tie-breaking does not affect the answer.

Rubric:

- 2 points for correct choice

- 4 points for correct explanation

iii. Uniform Cost Search (UCS)

Solution: YXYXY has the lowest cost (8) among the six goal states listed, so it will be returned by Uniform Cost Search. All of the other correct passwords have higher cost, so tie-breaking does not affect the answer.

Rubric:

- 2 points for correct choice
- 4 points for correct explanation

- b) [CO1, PO1] Assume that there is a single correct password of length 10, chosen uniformly at random from the state space. That means you do not know about the likelihood of the letters. The correct password is unknown, but a candidate password can be checked using your decryption software. Justify the statement: Given any heuristic, A* search will, on average, expand the same number of states as Depth-First Search.

Solution: Since we don't know the likelihood of the letters, all the letters have cost 1. The correct password is unknown, so the heuristic we define can't be a function of the correct password. Moreover, only full passwords can be checked (i.e. there is no way of measuring partial progress towards the goal.) We have to keep checking candidate passwords until we find the correct one. Since any password could have been chosen, all with equal probability, exploring the state space in any order will (on average) be equally effective.

Rubric:

- 2 points for correct assumption of the cost.
- 2 points for explaining ineffectiveness of the heuristic.
- 3 points for explanation regarding expanding up to the leaf nodes.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 1 Hour 30 Minutes

WINTER SEMESTER, 2020-2021
FULL MARKS: 75

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed.

There are **3 (three)** questions. Answer all **3 (three)** of them.

Figures in the right margin indicate marks of each question.

The square brackets on the start of each question denotes the corresponding CO and PO.

1. Consider that an agent is moving in the Gridworld shown in Figure ???. The only action available in cells *A*, *F*, *G*, and *H* is `Exit` with rewards $2X$, X , 0 , and 0 , respectively. Here, X = The last digit of your student ID + 1.

When the agent exits from a cell, it gets the reward for that cell as specified. From the other cells, the agent can only take the action `Left` or `Right`, which results in the agent moving to the immediate left or right cell, respectively.

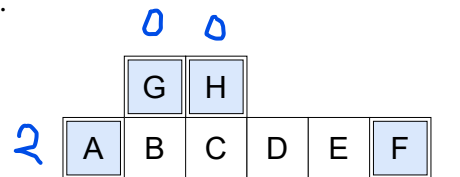


Figure 1: Gridworld for Question 1

If the agent is in cell *B* or *C*, and it takes a `Left` or `Right` action, it might fail sometimes. The agent will move in the desired direction with probability p , and it will fail and move up with probability $1 - p$. If the agent is in any other cell (*A*, *D*, *E*, *F*, *G*, or *H*), the action will always be successful.

Assume that the agent plays optimally, there is no living reward/penalty, and the discount factor is $\gamma \in [0, 1]$.

- a) [CO1, PO1] For each of the following policies, determine the value of each non-terminal state. 6 × 2

i. $\pi_{\text{Right}}(S) = \begin{cases} \text{Exit}, & \text{if } S \text{ is a Terminal State} \\ \text{Right}, & \text{otherwise} \end{cases}$

Solution:

- $B \rightarrow X\gamma^4 p^2$
- $C \rightarrow X\gamma^3 p$
- $D \rightarrow X\gamma^2$
- $E \rightarrow X\gamma$

(*A*, *D*, *E*, *F*, *G*, or *H*), the action will always be successful. Here we ignore $(1-p)$ as reward is zero. otherwise write that part also

Rubric:

- 0.5 points for each correct integer
- 0.5 points for each correct discount
- 0.5 points for each correct probability

$$\text{ii. } \pi_{\text{Left}}(S) = \begin{cases} \text{Exit,} & \text{if } S \text{ is a Terminal State} \\ \text{Left,} & \text{otherwise} \end{cases}$$

Solution:

$$\bullet B \rightarrow 2X\gamma p$$

$$\bullet C \rightarrow 2X\gamma^2 p^2$$

$$\bullet D \rightarrow 2X\gamma^3 p^2$$

$$\bullet E \rightarrow 2X\gamma^4 p^2$$

(A, D, E, F, G, or H), the action will always be successful. Here we ignore (1-p) as reward is zero. otherwise write that part also

Rubric:

- 0.5 points for each correct integer
- 0.5 points for each correct discount
- 0.5 points for each correct probability

b) [CO3, PO2, PO3] For what range of value for p will the agent choose π_{Left} over π_{Right} ?

7

Solution: For π_{Left} to be the optimal policy, for each of the four cells, π_{Left} needs to be greater than π_{Right} . Intuitively, the farther Right we are, the higher the value of moving Right, and the lower the value of moving Left (since the discount factor penalizes far-away rewards). Thus if moving Left is the optimal policy, we want to focus our attention on the Rightmost cell, E . We need:

$$\begin{aligned} \pi_{\text{Left}}(E) &> \pi_{\text{Right}}(E) \\ 2X\gamma^4 p^2 &> X\gamma \\ p &> \frac{1}{\sqrt{2\gamma^3}} \end{aligned}$$

Considering the probability range $[0, 1]$, we get: $p \in \left(\frac{1}{\sqrt{2\gamma^3}}, 1 \right]$.

However, $\lim_{\gamma \rightarrow 0} \frac{1}{\sqrt{2\gamma^3}} = \infty$. So the policy will not be optimal unless $\gamma > 0$.

Rubric:

- 2 points for describing the correct condition for π_{Left} being optimal.
- 2 points for the correct lower limit.
- 2 points for the correct upper limit.
- 1 point for the impossible condition.

c) [CO3, PO2, PO3] For what range of value for p will the agent choose π_{Right} over π_{Left} ?

6

Solution: Following the similar logic from the previous question, moving Right to be the

optimal policy, we want to focus our attention on the Leftmost cell, B . We get:

$$\pi_{\text{Right}}(B) > \pi_{\text{Left}}(B)$$

$$X\gamma^4 p^2 > 10\gamma p$$

$$p > \frac{2}{\gamma^3}$$

However, since γ ranges from 0 to 1, the right side of the expression ranges from 2 to ∞ , which means p (a probability, and thus bounded by 1) has no valid value. So the range is $\emptyset$.

Rubric:

- 2 points for describing the correct condition for π_{Right} being optimal.
- 2 points for the correct lower limit.
- 2 points for showing that it's not possible.

they are already over the time

2. a) [CO1, PO1] Two students, Rabbi and Sajal, are on their way to their class. The class is starting 8×2 right now, so both of them are running late. It takes 40 minutes to get to the classroom by walking, and only 10 minutes to get there by bus. Sajal prefers to go on a bus given the choice, so the bus incurs an additional utility bonus of 5. However, to get on the bus, they need to wait at the bus stand. Due to traffic congestion, the bus might come 20, 40, or 60 minutes, each with equal probability. For each of the following utility functions, determine whether that person should walk or take the bus:

+5 for
sajal
only

- i. Rabbi does not like being late. For him, the utility function for being late by time t is:

$$U_R(t) = \begin{cases} 0 & t \leq 0 \\ -2^{t/5} & t > 0 \end{cases}$$

Solution:

$$\begin{aligned} EU(\text{walk}) &= -2^{40/5} \\ &= -256 \end{aligned}$$

For rabbi, not sajal. So no bonus +5

$$\begin{aligned} EU(\text{bus}) &= \frac{1}{3} \left(-2^{(20+10)/5} - 2^{(40+10)/5} - 2^{(60+10)/5} \right) \\ &= -5824 \end{aligned}$$

Rabbi should walk.

Rubric:

- 3 points for expected utility of walking
- 3 points for expected utility of bus
- 2 point for the correct choice

- ii. Once Sajal is late, it does not matter to him how late he is. For him, the utility function for

being late by time t is:

$$U_S(t) = \begin{cases} 0 & t \leq 0 \\ -50 & t > 0 \end{cases}$$

Solution:

they are already over the time

$$EU(walk) = -50$$

$$EU(bus) = -50 + 5$$

$$= -45$$

Sajal should take the bus.

does not matter how late he is. So there is no t but still, the bonus +5 makes taking bus a better utility action

Rubric:

- 3 points for expected utility of walking
- 3 points for expected utility of bus
- 2 point for the correct choice

- b) [CO3, PO2, PO3] Assume that you have designed a deep learning model for determining whether a tomato leaf is infected by disease or not. Your system takes tomato leaf images as input and outputs “Positive” denoting that the leaf is infected, and “Negative” otherwise. After testing your model with a dataset, you have generated the confusion matrix to see how your model performs. You see that in 97% of the cases your model identifies a leaf as positive when the leaf is actually infected. And in 95% of the cases, your model identifies a leaf as negative, when the leaf is not infected. Suppose that, we know that 0.5% of the tomato leaf images in your dataset were infected by diseases. Explain why you should or should not trust the prediction of your system.

9

Solution: Let, I be the random variable denoting whether the leaf is infected or not. Let, P be the random variable denoting the output of the model.

Using Bayes’ rule, we can write: $P(p|i) = 97\%$ $P(n|i) = 3\%$ $P(n|-i) = 95\%$ $P(p|-i) = 5\%$
 $P(+i) = 0.5\%$, $P(-i) = 99.5\%$

$$\begin{aligned} P(+i|+p) &= \frac{P(+p|i)P(+i)}{P(+p)} \\ &= \frac{P(+p|i)P(+i)}{P(+p|i)P(+i) + P(+p|-i)P(-i)} \\ &= \frac{0.97 \times 0.005}{0.97 \times 0.005 + 0.05 \times 0.995} \\ &= 0.089 \end{aligned}$$

Given the model predicts a leaf to be infected, only 8.9% of the time the leaf will be actually infected! For example, if 1000 leaves are tested, it is expected that 995 of them are healthy leaves and 5 of them are infected by disease. From the 995 healthy leaves your model will predict $0.05 \times 995 \approx 50$ will be classified as “Positive”. From the 5 infected leaves, $0.95 \times 5 \approx 5$ will be classified as “Positive”. That means, out of the 55 positive results, only 5 are genuine.

Rubric:

- 2 points for the correct equation

- 4 points for the analysis
- 3 points for the conclusion
- Any correct argument (positive or negative) as conclusion will be accepted

3. [CO2, PO4] Consider that Pacman is moving in an unknown grid with only two cells X and Y . From each state, the Pacman can either move Left or Right. According to some policy π , Pacman got the following sequence of actions and rewards:

25

t	s_t	a_t	s_{t+1}	r_t
0	X	<i>Right</i>	Y	2
1	Y	<i>Right</i>	Y	-4
2	Y	<i>Left</i>	Y	0
3	Y	<i>Left</i>	X	3
4	X	<i>Left</i>	X	-1

Table 1: Samples table for Question 3

Model the scenario as Reinforcement Learning problem. Mention the states, actions, and Q-values for each state-action pair after performing sample-based Q-Learning. Recall the update function for sample-based Q-Learning is:

$$Q(s_t, a_t) \leftarrow (1 - \alpha)Q(s_t, a_t) + \alpha(r_t + \gamma \max_{a'} Q(s_{t+1}, a'))$$

Initially all Q-values are set to 0. The discount factor $\gamma = 0.5$ and the learning rate $\alpha = 0.5$.

Solution:

- States: $\{X, Y\}$

all Q initialized to 0 first

- Actions: $\{Left, Right\}$

At $t = 0$, we get:

$$\begin{aligned}
 Q(X, Right) &\leftarrow (1 - \alpha)Q(X, Right) + \alpha(r + \gamma \max_{a'} Q(Y, a')) \\
 &\leftarrow (1 - 0.5) \times 0 + 0.5 \times (2 + 0.5 \times 0) \\
 &\leftarrow 1
 \end{aligned}$$

At $t = 1$, we get:

$$\begin{aligned}
 Q(Y, Right) &\leftarrow (1 - \alpha)Q(Y, Right) + \alpha(r + \gamma \max_{a'} Q(Y, a')) \\
 &\leftarrow (1 - 0.5) \times 0 + 0.5 \times (-4 + 0.5 \times 0) \\
 &\leftarrow -2
 \end{aligned}$$

At $t = 2$, we get:

$$\begin{aligned} Q(Y, Left) &\leftarrow (1 - \alpha)Q(Y, Left) + \alpha(r + \gamma \max_{a'} Q(Y, a')) \\ &\leftarrow (1 - 0.5) \times 0 + 0.5 \times (0 + 0.5 \times 0) \quad \max(-2, 0, 0) = 0 \\ &\leftarrow 0 \end{aligned}$$

At $t = 3$, we get:


$$\begin{aligned} Q(Y, Left) &\leftarrow (1 - \alpha)Q(Y, Left) + \alpha(r + \gamma \max_{a'} Q(X, a')) \\ &\leftarrow (1 - 0.5) \times 0 + 0.5 \times (3 + 0.5 \times 1) \\ &\leftarrow \frac{7}{4} \quad \max(1, 0, 0) = 1 \end{aligned}$$

At $t = 4$, we get:

$$\begin{aligned} Q(X, Left) &\leftarrow (1 - \alpha)Q(X, Left) + \alpha(r + \gamma \max_{a'} Q(X, a')) \\ &\leftarrow (1 - 0.5) \times 0 + 0.5 \times (-1 + 0.5 \times 1) \\ &\leftarrow -\frac{1}{4} \quad \max(1, 0, 0) = 1 \end{aligned}$$

So the final values are:

- $Q(X, Left) = -\frac{1}{4}$
- $Q(X, Right) = 1$
- $Q(Y, Left) = \frac{7}{4}$
- $Q(Y, Right) = -2$

Rubric:

- 1 point for each item from states and actions
- 4 points for each update
- 0.25 points for each correct value of state-action pair